

Mu Jia

Ph.D. Candidate in Computer and Information Engineering
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RESEARCH INTERESTS

Statistical signal detection and estimation; radio-map construction and inference; wireless sensing and localization; integrated sensing and communication.

EDUCATION

- **The Chinese University of Hong Kong, Shenzhen** Shenzhen, China
Ph.D. Candidate in Computer and Information Engineering *Sep. 2023 – Present*
 - **Supervisors:** Prof. Junting Chen and Prof. Pooi-Yuen Kam
- **Harbin Institute of Technology, Shenzhen** Shenzhen, China
Master of Science in Information and Communication Engineering *2020 – Jan. 2023*
 - **Supervisor:** Prof. Tingting Zhang
- **University of Glasgow** Glasgow, UK / Chengdu, China
Bachelor of Engineering in Electrical and Electronic Engineering *2016 – 2020*
 - **GPA:** 17.93/22 (First-class Honours)
- **University of Electronic Science and Technology of China** Chengdu, China
Bachelor of Engineering in Electrical and Electronic Engineering *2016 – 2020*
 - **GPA:** 3.53/4.0 (Top 10%)

ACADEMIC EXPERIENCE

- **The Chinese University of Hong Kong, Shenzhen** Shenzhen, China
Teaching Assistant *2023 – 2026*
 - **Courses:** Honors Digital System Design (2026); Digital Communication (Spring/Fall 2025); Digital Signal Processing (Fall 2023, Fall 2024); Signals and Systems (Spring 2024)

HONORS AND AWARDS

- **Category II Scholarship, CUHK-Shenzhen:** 2023 – 2028
- **First Class Scholarship, HIT Shenzhen:** 2020 – 2021
- **Outstanding Student Scholarship, UESTC:** 2016 – 2020

PUBLICATIONS

Journal Articles and Submissions

- [1] **M. Jia**, J. Chen, Y.-C. Liang, and P.-Y. Kam, “Noncoherent detection of constant-envelope signals for mobile edge applications—asymptotically optimal detectors and a reliability-based intelligent detector,” *IEEE Trans. Mobile Comput.*, pp. 1–18, 2026, doi: 10.1109/TMC.2026.3730010.
- [2] **M. Jia**, T. Liang, T. Zhang, J. Chen, L. Zhou, T. Q. S. Quek, and P.-Y. Kam, “UAV detection and localization: A RF-based framework via multiple stations collaboration,” *IEEE Trans. Netw. Sci. Eng.*, vol. 13, pp. 5736–5751, 2026.
- [3] B. Li, H. Zhang, **M. Jia**, J. Chen, and N. Pappas, “Joint CFO-channel estimation under strong inter-cell interference for low-altitude radio mapping,” submitted to *IEEE Trans. Wireless Commun.*, 2026, arXiv:2512.01386.
- [4] R. Zhang, J. Yang, **M. Jia**, T. Zhang, and Y. Bao, “Resource optimization using a step-by-step scheme in wireless UWB sensing and localization networks,” *Appl. Sci.*, vol. 14, no. 6, Art. no. 2665, 2024.
- [5] M. Liu, **M. Jia**, and T. Zhang, “Scheduling in wireless cooperative localization networks with intelligent eavesdroppers,” *Wireless Commun. Mobile Comput.*, vol. 2023, Art. no. 7533455, 2023.

Conference Proceedings

- [6] **M. Jia**, H. Sun, J. Chen, and P.-Y. Kam, “Physics-aware tensor reconstruction for radio maps in pixel-based fluid antenna systems,” in *Proc. IEEE Int. Conf. Commun. Workshops (ICC Workshops)*, 2026, pp. 1–6.
- [7] **M. Jia**, J. Yang, and T. Zhang, “Power allocation in infrastructure-limited integrated sensing and localization wireless networks,” in *Proc. IEEE 96th Veh. Technol. Conf. (VTC2022-Fall)*, 2022, pp. 1–5.
- [8] **M. Jia**, S. Li, J. Le Kernec, S. Yang, F. Fioranelli, and O. Romain, “Human activity classification with radar signal processing and machine learning,” in *Proc. Int. Conf. UK-China Emerg. Technol. (UCET)*, 2020, pp. 1–5.
- [9] J. Tian, H. Wang, **M. Jia**, Y. Wang, J. Chen, and P.-Y. Kam, “Privacy-preserving federated radio map learning for wireless digital twins via adaptive noise allocation,” in the *2026 IEEE 27th Workshop on Signal Processing and Artificial Intelligence for Wireless Communications (SPAWC)*, Athens, Greece, Sep. 2026, Paper ID 241. Preprint: arXiv:2605.15560.
- [10] H. Sun, **M. Jia**, X. Yu, and J. Chen, “Bayesian localization of a moving source via map-informed sensing using a low-altitude UAV,” in *Proc. 30th Eur. Wireless Conf. (European Wireless)*, 2025, pp. 252–257.
- [11] Y. Zheng, **M. Jia**, and J. Chen, “Discovering CSI evolution using radio map for massive MIMO channel estimation,” in *Proc. IEEE Int. Conf. Commun. (ICC)*, 2024, pp. 5117–5122.
- [12] Y. Zheng, H. Sun, B. Li, **M. Jia**, and J. Chen, “A radio map approach for antenna pattern calibration in low-altitude air-to-ground sensing,” in *Proc. IEEE/CIC Int. Conf. Commun. China Workshops (ICCC Workshops)*, 2024, pp. 84–89.
- [13] X. Wang, **M. Jia**, X. Meng, and T. Zhang, “Multipath ghosts mitigation for radar-based positioning systems,” in *Proc. IEEE 96th Veh. Technol. Conf. (VTC2022-Fall)*, 2022, pp. 1–6.
- [14] B. Gao, **M. Jia**, T. Zhang, and Q. Zhang, “Reliable target positioning in complicated environments using multiple radar observations,” in *Proc. IEEE Global Commun. Conf. (GLOBECOM)*, 2021, pp. 1–6.
- [15] S. Li, **M. Jia**, J. Le Kernec, S. Yang, F. Fioranelli, and O. Romain, “Elderly care: Using deep learning for multi-domain activity classification,” in *Proc. Int. Conf. UK-China Emerg. Technol. (UCET)*, 2020, pp. 1–4.